CSE 464 Project Part 1 - DOT Graph Parser (Java + Maven)

Name: Devendra Janyani
ASU ID: 1225359265
GitHub Repo: https://github.com/devjanyani/CSE-464-2026-djanyani

Project Overview:

This project parses directed graphs written in Graphviz DOT format, constructs an internal directed graph representation, and supports graph manipulation and export. The program can: (1) parse a DOT file into a Graph object, (2) add nodes and edges, (3) output the graph as DOT and as a readable text summary, and (4) export a PNG visualization using Graphviz. The project is built with Maven and includes JUnit tests for the implemented features.

Environment Setup:

The project was developed and tested with the following tools:

1. Java: JDK 25
2. Build System: Maven
3. IDE: IntelliJ IDEA Community Edition
4. Graph Visualization: Graphviz

Build Instructions:

The project is built using Maven. The following commands compile the source code, run all unit tests, and package the project into a runnable JAR file.

```
cd graph-parser

mvn package
```

Screenshot: Successful Maven build using mvn package.:

```
PS C:\Users\dejan\IdeaProjects\CSE-464-2026-djanyani\graph-parser> mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< edu.asu.cse464:graph-parser >--------------------
[INFO] Building graph-parser 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ graph-parser ---
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ graph-parser ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ graph-parser ---
[INFO] skip non existing resourceDirectory C:\Users\dejan\IdeaProjects\CSE-464-2026-djanyani\graph-parser\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ graph-parser ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ graph-parser ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running edu.asu.cse464.DotParserTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.072 s -- in edu.asu.cse464.DotParserTest
[INFO] Running edu.asu.cse464.GraphTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in edu.asu.cse464.GraphTest
[INFO] Running edu.asu.cse464.GraphvizExportTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.632 s -- in edu.asu.cse464.GraphvizExportTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ graph-parser ---
[INFO] Building jar: C:\Users\dejan\IdeaProjects\CSE-464-2026-djanyani\graph-parser\target\graph-parser-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.988 s
[INFO] Finished at: 2026-03-04T20:16:52-07:00
[INFO] ------------------------------------------------------------------------

PS C:\Users\dejan\IdeaProjects\CSE-464-2026-djanyani\graph-parser>
```

Running the application

```
java -cp target/graph-parser-1.0-SNAPSHOT.jar edu.asu.cse464.Main
```

Screenshot: Console output after running the application:

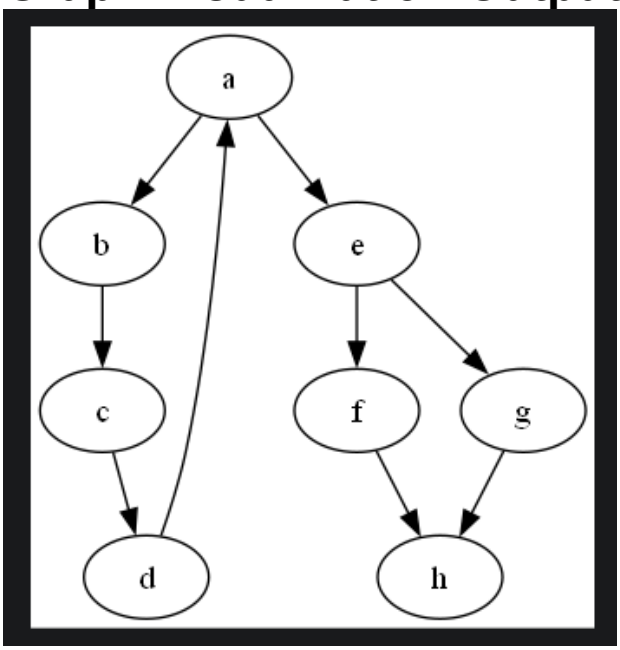
```
Terminal   Local

PS C:\Users\dejan\IdeaProjects\CSE-464-2026-djanyani\graph-parser> java -cp target/graph-parser-1.0-SNAPSHOT.jar edu.asu.cse464.Main
Number of nodes: 8
Nodes: [a, b, c, d, e, f, g, h]
Number of edges: 9
Edges:
a -> b
a -> e
b -> c
c -> d
d -> a
e -> f
e -> g
f -> h
g -> h
Wrote DOT to: output.dot
Wrote graph text to: output.txt
Wrote PNG to: output.png

PS C:\Users\dejan\IdeaProjects\CSE-464-2026-djanyani\graph-parser>
```

Example Input (DOT file):

sample.dot

```
digraph {
  a -> b;
  b -> c;
  c -> d;
  d -> a;
  a -> e;
  e -> f;
  e -> g;
  f -> h;
  g -> h;
}
```

Graph Visualization Output:



Unit Tests:

Command to Run Tests:

mvn test

GraphTest:

Tests node creation, edge creation, and graph output functionality.

DotParserTest:

Tests that DOT files are parsed correctly and verifies isolated node parsing.

GraphvizExportTest:

Tests that the graph can be exported as a PNG image using Graphviz.

```
PS C:\Users\dejan\IdeaProjects\CSE-464-2026-djanyani\graph-parser> mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< edu.asu.cse464:graph-parser >--------------------
[INFO] Building graph-parser 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ graph-parser ---
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ graph-parser ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ graph-parser ---
[INFO] skip non existing resourceDirectory C:\Users\dejan\IdeaProjects\CSE-464-2026-djanyani\graph-parser\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ graph-parser ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ graph-parser ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running edu.asu.cse464.DotParserTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.092 s -- in edu.asu.cse464.DotParserTest
[INFO] Running edu.asu.cse464.GraphTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in edu.asu.cse464.GraphTest
[INFO] Running edu.asu.cse464.GraphvizExportTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.794 s -- in edu.asu.cse464.GraphvizExportTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.969 s
[INFO] Finished at: 2026-03-04T20:24:17-07:00
[INFO] ------------------------------------------------------------------------
```

Git pushes:

Feature: Graph Data Structure
Commit: https://github.com/devjanyani/CSE-464-2026-djanyani/commit/f91d6a1

Feature: DOT Graph Parser
Commit: https://github.com/devjanyani/CSE-464-2026-djanyani/commit/b522520

Feature: DOT Output Serialization
Commit: https://github.com/devjanyani/CSE-464-2026-djanyani/commit/7a329ab

Feature: Graphviz PNG Export
Commit: https://github.com/devjanyani/CSE-464-2026-djanyani/commit/e6aa8cf

Unit Tests:

Commit: **https://github.com/devjanyani/CSE-464-2026-djanyani/commit/15d7dd4**

Project Structure:

```
graph-parser/
    src/main/java/edu/asu/cse464
        Graph.java
        DotParser.java
        Main.java
    src/test/java/edu/asu/cse464
        GraphTest.java
        DotParserTest.java
        GraphvizExportTest.java
    input/sample.dot
    pom.xml
```

Build instructions:

```
cd graph-parser
mvn clean package
```